De
UW

Department of Computer Science

Assignment # 01

Course Code: CS-121
Course Title: Digital Logic Design
Total Marks: 20

Discipline/Semester: BSCS- 2nd B
Date/Deadline: 27h March 2025
Assigned Date: 25h March 2025.

CLO-1: **Identify and explain** fundamental concepts of digital logic design including basic and universal gates, number systems, binary coded systems, basic concepts of digital system.
CLO-2: **Acquired knowledge** to apply techniques related to the design and analyze of digital electronic circuits including Boolean algebra.

CLO2-GA2-BT-Level C2

Q1: Explain the answers to all questions in detail.

a) **Convert** Gray code to binary code
[5]

i. **110010**
ii. **1011001**

b) **Explain** the given below question for this Boolean function in detail. Identify and illustrate the steps involved in its simplification.
[10]

$$F = xy'z + x'y'z + w'xy + wx'y + wxy$$

(a) Obtain the truth table of ***F***.

(b) Draw the logic diagram, using the original Boolean expression.

(c) Use Boolean algebra to simplify the function to a minimum number of literals.

(d) Obtain the truth table of the function from the simplified expression and show that it is the same as the one in part (a).

(e) Draw the logic diagram from the simplified expression, and compare the total number of gates with the diagram of part (b)

c) **Describe** the Boolean expression (in Sum of Products form SOP) for the logic circuit that will have a 1 output when X = 0, Y = 0, Z = 1 and X = 1, Y = 1, Z = 0, and a zero (0) output for all other input states. Draw the logic diagram for this circuit.
[5]